BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 221 · A*: heuristic reliability and route evaluation

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Calculate f when g=4 and h=3.

Q2. What is the role of a predecessor table?

Q3. If S–A=2 and A–G=5, what is this route's total cost?

Q4. The true cheapest remaining cost from A is 4. Is h(A)=6 admissible?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

admissible — Heuristic never overestimates the true cheapest remaining cost.

consistent — Heuristic obeys h(u) ≤ edge(u,v)+h(v) for every edge.

suboptimal — A valid route whose cost is greater than the cheapest possible.

frontier — Discovered vertices awaiting expansion.

heuristic — An estimated remaining cost used to guide search.

New learning

Find the real remaining cost — From A, the cheapest route A→B→G costs 2+2=4. Therefore h(A)=3 is admissible, but h(A)=9 is not.

Safe estimates support optimal search — With h(A)=3, A is expanded before B, improving the route to B and finally G at cost 6.

Watch the misleading estimate — With h(A)=9, A has f=11; B has f=7 and leads to goal at cost 8.

Why the error matters — A* that stops on the first goal removed from the open set can return this 8-cost path before exploring A, even though a 6-cost route exists.

Check the guarantee — Admissibility means no overestimate. Consistency adds the edge test h(u) ≤ weight(u,v) + h(v). Without reopening settled nodes, A* requires consistency for its shortest-path guarantee. Explain this as well as the route cost.

Worked example

Check admissibility and trace U→G with safe estimates.

1. Predict the choice — V has g1,h3,f4; W has g5,h1,f6. Expand V.

2. Review the choice — V improves W to g3 and f4; expand W.

3. Review the choice — W improves the goal G to g5,h0,f5.

4. Review the choice — The path is U→V→W→G with total cost 5; the heuristics do not exceed the actual remaining costs.

We Do · Practise with guidance

Can a bad estimate make A* return a longer route?

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Edges M–N=2, M–O=5, N–O=2, N–G=7, O–G=2.
h(M)=4,h(N)=7 (unsafe),h(O)=1,h(G)=0.
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Predict the choice

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

After M, O has f5+1=6; N has f2+7=9, so O is selected.

Review the choice

O reaches G at g7,f7. G is selected before N, so the early-goal algorithm returns cost 7.

But M→N→O→G costs 2+2+2=6; the bad h(N)=7 overestimates N's true remaining cost 4.

Using an admissible estimate such as h(N)=3 allows the cheaper path to be found.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

The true remaining distance from node A to goal G is 6. Decide whether h(A)=4 and h(A)=8 are admissible, with reasons.

Question 2

Edges S–A=2, S–B=6, A–B=2, B–G=2, A–G=8. Calculate the best S→G route and explain why h(A)=9 can mislead A*.

Question 3

Compare Dijkstra's algorithm and A* in their selection rule, heuristic use and guarantee of shortest paths.

Exit Ticket · Answer before checking

If a heuristic overestimates the true remaining cost, A* may lose its shortest-route guarantee.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. f=7.

Q2. It lets us reconstruct the path by following predecessors backwards.

Q3. 7.

Q4. No. It overestimates the true cost.

You Do pages · Independent questions

Q1. [4] h(A)=4 is admissible [1] because it does not exceed 6 [1]; h(A)=8 is inadmissible [1] because it overestimates 6 [1].

Q2. [6] Route S→A→B→G [2], cost 6 [1]; from A the true cheapest remaining cost is 4 [1], so h(A)=9 overestimates [1] and can delay expanding A while a more expensive goal route appears [1].

Q3. [6] Dijkstra selects lowest tentative g [1] and does not need h [1]; A* selects lowest f=g+h [2]; Dijkstra needs non-negative edges [1]; A* requires suitable admissible heuristic (and appropriate closed-set handling, usually consistency) to guarantee optimality [1].

Exit Ticket · Model answer

True. An inadmissible estimate can make a less costly path appear too unattractive to explore before stopping.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding